



**UNIVERSITI
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WIA 1007

INTRODUCTION TO DATA SCIENCE

PROJECT REPORT

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Introduction

In the modern banking sector, data-driven approaches have become essential for enhancing decision-making processes and achieving operational efficiency. One critical area where data science can make a significant impact is in optimizing marketing campaigns for financial products. This project focuses on predicting customer subscriptions to term deposits using a detailed dataset that includes customer demographics, financial attributes, and past interactions with marketing campaigns. By applying machine learning techniques, this study aims to uncover key patterns in customer behaviour and develop predictive models that enable banks to improve the precision and effectiveness of their marketing strategies.

Background of The Problem

Marketing campaigns are a cornerstone of customer acquisition in the banking industry, but they often incur high costs with uncertain returns. Many campaigns are broadly targeted, leading to wasted resources and suboptimal engagement. Identifying customers who are more likely to subscribe to financial products, such as term deposits, can significantly enhance the efficiency and impact of these campaigns.

However, predicting customer behaviour in this context presents challenges due to the wide variety of customer profiles, preferences, and behaviours influenced by demographic, financial, and historical factors. This complexity requires robust analytical methods to extract actionable insights from large datasets.

This project addresses these challenges by leveraging data science to analyse patterns and trends in customer data. The results provide banks with tools to improve customer targeting, reduce marketing costs, and foster better customer relationships. By focusing on the factors that drive subscription behaviour, this study demonstrates the potential of data-driven approaches to transform marketing strategies in the banking sector.

Data Preprocessing

Table 1. Data Properties 1: Types of Data and Data Types

Variable	Type of data	Data types	Measurement level	Units	Range	Min value	Top value	Unique values	Null values	Outliers
age	Numeric	int64	ratio	years	18 - 72	18	32	77	0	Yes, <18 or >72
job	Categorical	string	ratio	-	blue-collar, management, technician, admin, etc.	unknown	blue-collar	12	0	No
marital	Categorical	string	nominal	-	married, single, divorced	divorced	married	3	0	No
education	Categorical	string	nominal	-	primary, secondary, tertiary, unknown	unknown	secondary	4	0	No
default	Categorical	string	ordinal	-	yes, no	yes	no	2	0	No
balance	Numerical	float64	nominal	euros	-1132 - 1972	-1129	0	3196	0	Yes, <-1132 or >1972
housing	Categorical	string	nominal	-	yes, no	no	yes	2	0	No
loan	Categorical	string	nominal	-	yes, no	yes	no	2	0	No
contact	Categorical	string	nominal	-	cellular, telephone, unknown	telephone	cellular	3	0	No
day	Numerical	int64	interval	days	1 - 31	1	20	31	0	Yes, <1 or >31
month	Categorical	string	nominal	-	jan, feb, march, etc.	dec	may	12	0	No
duration	Numerical	int64	ratio	seconds	0 - 634.5	0	90	1505	0	Yes, <0 or >634.5

campaign	Numerical	int64	ratio	-	1 - 6	1	1	47	0	Yes, <1 or >6
pdays	Numerical	int64	ratio	-	-1	-1	-1	531	0	No
previous	Numerical	int64	ratio	-	0 - 20	0	0	39	0	Yes, >20
outcome	Categorical	string	nominal	-	success, failure, other, unknown	success	unknown	4	0	No
year	Categorical	string	nominal	-	yes, no	yes	no	2	0	No
duration campaign ratio	Numerical	float64	ratio	seconds	0 - 241	0	40	4593	0	Yes, <0 or >241
age group	Categorical	string	ordinal	-	youth, middle-aged, senior, elder	elder	middle-aged	4	0	No
balance smoothed	Numerical	float64	ratio	euros	-598 - 1548	-576	0	5492	0	Yes, <-598 or >1548

Table 2. Data Properties: Statistics

Variable	Frequency	Percentile (25th, 50th, 75th)	Data Completeness	Mean	Median	Mode	Standard Deviation	Variance	Skewness	Kurtosis
age	-	[32.0, 39.0, 48.0]	100.00%	40.63	39	32	10.47	109.55	0.69	0.33
job	{'blue-collar': '22.11%', 'management': '20.10%', 'technician': '16.93%', 'admin.': '11.67%', 'services': '9.49%', 'retired': '4.73%', 'self-	-	100.00%	-	-	blue-collar	-	-	-	-

	'11.52%', 'nov': '7.59%', 'apr': '6.41%', 'feb': '5.96%', 'jan': '3.26%', 'oct': '1.49%', 'sep': '1.22%', 'mar': '0.97%', 'dec': '0.44%'} 									
duration	-	[103.0, 179.0, 316.0]	100.00%	256.1 6	179	124	255.5	65282.42	3.13	17.13
campaign	-	[1.0, 2.0, 3.0]	100.00%	2.77	2	1	3.12	9.71	4.83	37.22
pdays	-	[-1.0, - 1.0, -1.0]	100.00%	40.3	-1	-1	101.14	10229.31	2.62	6.88
previous	-	[0.0, 0.0, 0.0]	100.00%	0.57	0	0	2.32	5.37	45.24	4912.32
poutcome	{'unknown': '82.06%', 'failure': '10.71%', 'other': '4.05%', 'success': '3.18%'}	-	100.00%	-	-	unknown	-	-	-	-
y	{'no': '88.83%', 'yes': '11.17%'}	-	100.00%	-	-	no	-	-	-	-
duration_campaign_ratio	-	[30.25, 60.85, 115.48]	100.00%	91.04	60.85	45	100.3	10059.19	3.38	21.4
age_group	{'Middle-aged': '76.53%', 'Senior': '20.99%', 'Youth': '1.89%', 'Elder': '0.59%'}	-	100.00%	-	-	Middle-aged	-	-	-	-
balance_smoothed	-	[243.33, 547.5, 960.0]	100.00%	640.6 3	547.5	0	517.84	268162.0 5	0.8	0.55

Exploratory Data Analysis (EDA)

1. What is the relationship between age and subscription rates?

Middle-aged group shows the highest potential for term deposit subscriptions. While youth and elder might have lower subscription rates.

2. How do subscription rates vary across job categories?

Blue-collar workers subscription rate might be lower compared to management or technicians. Students and Retirees might have lower subscription rates due to limited income or different financial priorities.

3. What is the impact of marital status on subscription rates?

Married individuals form the largest segment and could be a primary target audience for term deposits. Their financial stability and family-oriented goals might make them more receptive. While single individuals' subscription rate might be lower than married individuals but higher than divorced individuals. This could be due to individual financial planning preferences.

Machine Learning Models

Machine learning models are algorithms that learn patterns from data to make predictions or decisions. For this project, we implemented and evaluated two models: **Logistic Regression** and **Random Forest Classifier**. Each model was chosen for its unique strengths in handling classification problems with mixed data types.

1. Logistic Regression

- **Type:** Linear Model (Shallow Learning).
- **Implementation:** Built using Logistic Regression from scikit-learn. Includes a preprocessing pipeline for OneHotEncoding categorical features. Limited to linear decision boundaries.
- **Training:** Fit to the training data after OneHotEncoding the categorical variables. Used a maximum of 1000 iterations to ensure convergence.
- **Performance:** Evaluated on test data with metrics such as accuracy, precision, recall, and F1 score.

2. Random Forest Classifier

- **Type:** Ensemble Learning (Tree-Based).
- **Implementation:** Built using Random Forest Classifier from scikit-learn. Combined with OneHotEncoding for preprocessing. Uses multiple decision trees and combines their outputs for final predictions.
- **Training:** Fit to the training data with default hyperparameters and a fixed random seed.
- **Performance:** Evaluated on both training and test datasets, with accuracy as the primary metric.

Results

Accuracy comparison of models:

- Logistic Regression: 90.90%
- Random Forest: 91.28%

Performance comparison of models:

- Logistic Regression: Moderate accuracy, good interpretability.
- Random Forest: Best accuracy, robust performance.

Model	Strengths	Weaknesses
Logistic Regression	Simple, interpretable	Limited to linear relationships
Random Forest	Handles non-linear data, reduces overfitting	Less interpretable, slower for large datasets

Conclusion

In conclusion, the analysis of the bank marketing dataset reveals that leveraging machine learning can significantly enhance predictive capabilities regarding customer behaviour. The Random Forest model stands out as the best choice due to its superior accuracy and ability to handle non-linear relationships effectively. By understanding customer through EDA, banks can tailor their marketing strategies to improve subscription rates for term deposits. Overall, this analysis provides a solid foundation for using data-driven insights in banking to support improved decision-making and strategic planning.